Alex Kalinichenko

Senior / Lead Software Engineer

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PROFILE

Senior / Lead Software Engineer with 8+ years shipping production web applications end to end, and 5 years focused on backend, data and AI systems. Currently building a real-time crowd-analytics platform for live events on Google Cloud, owning the React/TypeScript app, the NestJS API and the Python and BigQuery data pipelines through to production. Previously led a team of 6 building a conversational AI platform, from architecture and LLM/RAG pipelines to on-premises enterprise deployment.

TECHNICAL SKILLS

Languages: TypeScript / JavaScript (8y) · Node.js / NestJS (5y) · Python (2y) · SQL

Frontend: React, Next.js, Vite, Redux / Redux-Saga, Jotai, React Query, GraphQL, WebSockets, Formik · Material-UI, Radix UI, Tailwind, styled-components · Mapbox, TerraDraw, Recharts · Storybook, Jest, Playwright, Figma

Backend: Node.js, NestJS, Express · REST, GraphQL, WebSockets, JWT, RBAC · RxJS · Web scraping & automation (Puppeteer, Cheerio) · Document processing (pdf-lib, PDFKit, pdf2pic, jszip, csv-parser)

Python & Data: FastAPI, Pydantic, pandas, PyTorch · BigQuery, Pub/Sub, Dataflow (Apache Beam) · streaming and batch pipelines

AI / LLM: LLM integration, RAG pipelines, AI OCR, document analysis & extraction, on-prem model serving

Infrastructure: Google Cloud (Cloud Run, Cloud SQL, Compute Engine, Cloud Build, Secret Manager, Memorystore), AWS (Lambda), Hetzner · Terraform, Docker, GitLab CI/CD · Cloudflare (DNS, Tunnels) · Redis, RabbitMQ · GPU services, on-premises delivery

Databases: PostgreSQL / PostGIS (TypeORM, Prisma), row-level security, partitioning · MongoDB (Mongoose)

Observability: Sentry, Datadog, Splunk, Coralogix, Cloud Logging & Monitoring, OpenReplay, FullStory, Microsoft Clarity

EXPERIENCE

Senior Software Engineer — Metamindz

Nov 2025 – Present

Stack: *TypeScript, React, Vite, Tailwind, Mapbox, TerraDraw, NestJS, TypeORM, PostgreSQL / PostGIS, Python, BigQuery, Pub/Sub, Dataflow, Cloud Run, Cloud SQL, Terraform, Cloudflare, Playwright*

- Full-stack engineer on a real-time crowd-analytics platform for live events and venues, turning Wi-Fi/BLE sensor, wearable and phone-SDK signals into live occupancy, density heatmaps, footfall and dwell analytics.
- Own the operator web app, the NestJS API and the GCP data pipelines end to end, from ticket to production rollout and verification, working directly with the product owner.
- Built analytics versioning: every reprocess creates a separate, promotable version, so customer-facing numbers change only once approved; rolled out to production with migrations and row-level security.
- Diagnosed a production database pinned at 100% CPU by a security policy forcing full scans of a 25 GB table, and replaced it with a trigger that brought CPU back to ~10%.
- Moved the sensor data stream behind an internal load balancer and a Terraform-managed Cloudflare tunnel, cutting billed external ingress from 120–190 GB a day to zero.
- Built a wearable ingest gateway on Cloud Run with Pub/Sub and a BigQuery archive, and a positioning pipeline for the mobile SDK.
- Built the venue deployment planning tool on Mapbox and TerraDraw, plus a dashboard builder, custom metrics, PDF area reports, gate counts and app analytics.
- Stood up a dev environment at parity with production, including importing production events and replaying production data.

Lead Software Engineer — Metadialog

Sep 2021 – Nov 2025

Stack: *Python, React, Next.js, PostgreSQL, TypeORM / SQLAlchemy, Redis, RabbitMQ, Docker, WebSockets, LLM/RAG*

- Led a team of 6 engineers, owning delivery end to end — sprint planning, demos, and syncs with C-level stakeholders.

- Architected and built a conversational AI platform from scratch that lets customers train and deploy their own bots, with a full LLM + RAG request pipeline (UI → API → LLM → RAG → LLM → UI).
- Owned the entire infrastructure stack: React frontend, NestJS backend, and GPU-deployed Python services across GCP and Hetzner, with Cloudflare DNS and automated GitLab/Docker deployments.
- Delivered on-premises deployments onto customer-provided servers for enterprise clients with strict data-residency requirements.
- Built integrations with major support and communication systems — Freshdesk, Zendesk, Freshchat, Microsoft Teams, Outlook, and Gmail.
- Led technical R&D for AI-driven features, including PDF editing with AI, automated questionnaire completion from documents, and AI OCR.
- Ran technical discovery and built POCs on sales/demo calls to convert prospective enterprise customers.

Senior Software Engineer — Ciklum *Dec 2020 – Sep 2021*

Stack: *React, Redux, Node.js, Express, Angular, TypeScript*

- Designed the architecture for a high-load application service.
- Integrated a React application into an existing Angular app as a submodule, contributing across both codebases.
- Collaborated with a distributed Angular team abroad, contributing directly to their modules.
- Interviewed and assessed mid-level engineering candidates.

Frontend Developer — Diceus *Dec 2018 – Dec 2020*

Stack: *React, Redux, Redux-Saga, Formik, pdfmake, Jest, lodash, react-router-dom, Tailwind, WebSockets, Mapbox, styled-components, TypeScript*

- Delivered features within a large frontend team (~10 engineers) and mentored junior developers.
- Took ownership of a backend feature on the frontend side, cutting 2 weeks off the original delivery estimate.
- Translated customer requirements into design specs, bridging customer, design, and engineering.
- Hardened business-critical features with a comprehensive test suite.
- Ramped quickly on new technologies to support pre-sales and technical demos.

Mobile Developer (Freelance) *Apr 2020 – Oct 2020*

Stack: *React Native, Redux, react-native-gl, react-native-ffmpeg, react-native-ble-manager, Firebase, TypeScript*

- Built a React Native app to stream images and GIFs to an LED panel over Bluetooth.
- Designed a data-hashing approach that improved transmission efficiency.
- Optimized an in-app editor for drawing and composing GIFs.

Frontend Developer — Merehead *Feb 2018 – Nov 2018*

Stack: *React, Redux, Redux-Thunk, Redux-Saga, react-ga, CSS, SCSS, Material-UI, Jest, WebSockets*

- Built a large-scale social network application alongside PM, design, QA, and startup leadership.
- Rapidly resolved critical issues across legacy projects.
- Integrated Google Analytics for the company's largest clients.
- Customized an audio player in a music application to match design specs.

EDUCATION & LANGUAGES

M.Sc. Cybersecurity — State University of Telecommunications *2016 – 2022*

Languages: English (Advanced) · Ukrainian (Native) · Russian (Native)